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CSES Problem Set**Subordinates**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given the structure of a company, your task is to calculate for each employee the number of their subordinates.

Input

The first input line has an integer n : the number of employees. The employees are numbered $1, 2, \dots, n$, and employee 1 is the general director of the company.

After this, there are $n - 1$ integers: for each employee $2, 3, \dots, n$ their direct boss in the company.

Output

Print n integers: for each employee $1, 2, \dots, n$ the number of their subordinates.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input:

```
5
1 1 2 3
```

Output:

```
4 1 1 0 0
```

CSES Problem Set

Tree Matching

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a tree consisting of n nodes.

A *matching* is a set of edges where each node is an endpoint of at most one edge. What is the maximum number of edges in a matching?

Input

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output

Print one integer: the maximum number of pairs.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5
1 2
1 3
3 4
3 5
```

Output:

```
2
```

Explanation: One possible matching is $(1, 2)$ and $(3, 4)$.

CSES Problem Set

Tree Diameter

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a tree consisting of n nodes.

The *diameter* of a tree is the maximum distance between two nodes. Your task is to determine the diameter of the tree.

Input

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output

Print one integer: the diameter of the tree.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5
1 2
1 3
3 4
3 5
```

Output:

```
3
```

Explanation: The diameter corresponds to the path $2 \rightarrow 1 \rightarrow 3 \rightarrow 5$.

CSES Problem Set

Tree Distances I

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a tree consisting of n nodes.

Your task is to determine for each node the maximum distance to another node.

Input

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output

Print n integers: for each node $1, 2, \dots, n$, the maximum distance to another node.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5
1 2
1 3
3 4
3 5
```

Output:

```
2 3 2 3 3
```

CSES Problem Set

Tree Distances II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a tree consisting of n nodes.

Your task is to determine for each node the sum of the distances from the node to all other nodes.

Input

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output

Print n integers: for each node $1, 2, \dots, n$, the sum of the distances.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5
1 2
1 3
3 4
3 5
```

Output:

```
6 9 5 8 8
```

CSES Problem Set

Company Queries I

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A company has n employees, who form a tree hierarchy where each employee has a boss, except for the general director.

Your task is to process q queries of the form: who is employee x 's boss k levels higher up in the hierarchy?

Input

The first input line has two integers n and q : the number of employees and queries. The employees are numbered $1, 2, \dots, n$, and employee 1 is the general director.

The next line has $n - 1$ integers $e_2, e_3, \dots, e_n$: for each employee $2, 3, \dots, n$ their boss.

Finally, there are q lines describing the queries. Each line has two integers x and k : who is employee x 's boss k levels higher up?

Output

Print the answer for each query. If such a boss does not exist, print -1 .

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq e_i \leq i - 1$
- $1 \leq x \leq n$
- $1 \leq k \leq n$

Example

Input:

```
5 3
1 1 3 3
4 1
4 2
4 3
```

Output:

```
3
1
-1
```

CSES Problem Set**Company Queries II**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A company has n employees, who form a tree hierarchy where each employee has a boss, except for the general director.

Your task is to process q queries of the form: who is the lowest common boss of employees a and b in the hierarchy?

Input

The first input line has two integers n and q : the number of employees and queries. The employees are numbered $1, 2, \dots, n$, and employee 1 is the general director.

The next line has $n - 1$ integers $e_2, e_3, \dots, e_n$: for each employee $2, 3, \dots, n$ their boss.

Finally, there are q lines describing the queries. Each line has two integers a and b : who is the lowest common boss of employees a and b ?

Output

Print the answer for each query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq e_i \leq i - 1$
- $1 \leq a, b \leq n$

Example

Input:

```
5 3
1 1 3 3
4 5
2 5
1 4
```

Output:

```
3
1
1
```


CSES Problem Set**Distance Queries**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a tree consisting of n nodes.

Your task is to process q queries of the form: what is the distance between nodes a and b ?

Input

The first input line contains two integers n and q : the number of nodes and queries. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Finally, there are q lines describing the queries. Each line contains two integers a and b : what is the distance between nodes a and b ?

Output

Print q integers: the answer to each query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 3
1 2
1 3
3 4
3 5
1 3
2 5
1 4
```

Output:

```
1
3
2
```

CSES Problem Set

Counting Paths

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a tree consisting of n nodes, and m paths in the tree.

Your task is to calculate for each node the number of paths containing that node.

Input

The first input line contains integers n and m : the number of nodes and paths. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Finally, there are m lines describing the paths. Each line contains two integers a and b : there is a path between nodes a and b .

Output

Print n integers: for each node $1, 2, \dots, n$, the number of paths containing that node.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 3
1 2
1 3
3 4
3 5
1 3
2 5
1 4
```

Output:

```
3 1 3 1 1
```

CSES Problem Set

Subtree Queries

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a rooted tree consisting of n nodes. The nodes are numbered $1, 2, \dots, n$, and node 1 is the root. Each node has a value.

Your task is to process following types of queries:

1. change the value of node s to x
2. calculate the sum of values in the subtree of node s

Input

The first input line contains two integers n and q : the number of nodes and queries. The nodes are numbered $1, 2, \dots, n$.

The next line has n integers $v_1, v_2, \dots, v_n$: the value of each node.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Finally, there are q lines describing the queries. Each query is either of the form "1 s x " or "2 s ".

Output

Print the answer to each query of type 2.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq a, b, s \leq n$
- $1 \leq v_i, x \leq 10^9$

Example

Input:

```
5 3
4 2 5 2 1
1 2
1 3
3 4
3 5
2 3
1 5 3
2 3
```

Output:

```
8
10
```

CSES Problem Set

Path Queries

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a rooted tree consisting of n nodes. The nodes are numbered $1, 2, \dots, n$, and node 1 is the root. Each node has a value.

Your task is to process following types of queries:

1. change the value of node s to x
2. calculate the sum of values on the path from the root to node s

Input

The first input line contains two integers n and q : the number of nodes and queries. The nodes are numbered $1, 2, \dots, n$.

The next line has n integers $v_1, v_2, \dots, v_n$: the value of each node.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Finally, there are q lines describing the queries. Each query is either of the form "1 s x " or "2 s ".

Output

Print the answer to each query of type 2.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq a, b, s \leq n$
- $1 \leq v_i, x \leq 10^9$

Example

Input:

```
5 3
4 2 5 2 1
1 2
1 3
3 4
3 5
2 4
1 3 2
2 4
```

Output:

```
11
8
```

CSES Problem Set

Path Queries II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a tree consisting of n nodes. The nodes are numbered $1, 2, \dots, n$. Each node has a value.

Your task is to process following types of queries:

1. change the value of node s to x
2. find the maximum value on the path between nodes a and b .

Input

The first input line contains two integers n and q : the number of nodes and queries. The nodes are numbered $1, 2, \dots, n$.

The next line has n integers $v_1, v_2, \dots, v_n$: the value of each node.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Finally, there are q lines describing the queries. Each query is either of the form "1 s x " or "2 a b ".

Output

Print the answer to each query of type 2.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq a, b, s \leq n$
- $1 \leq v_i, x \leq 10^9$

Example

Input:

```
5 3
2 4 1 3 3
1 2
1 3
2 4
2 5
2 3 5
1 2 2
2 3 5
```

Output:

```
4 3
```

CSES Problem Set

Distinct Colors

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a rooted tree consisting of n nodes. The nodes are numbered $1, 2, \dots, n$, and node 1 is the root. Each node has a color.

Your task is to determine for each node the number of distinct colors in the subtree of the node.

Input

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

The next line consists of n integers $c_1, c_2, \dots, c_n$: the color of each node.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output

Print n integers: for each node $1, 2, \dots, n$, the number of distinct colors.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$
- $1 \leq c_i \leq 10^9$

Example

Input:

```
5
2 3 2 2 1
1 2
1 3
3 4
3 5
```

Output:

```
3 1 2 1 1
```

CSES Problem Set**Finding a Centroid**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given a tree of n nodes, your task is to find a *centroid*, i.e., a node such that when it is appointed the root of the tree, each subtree has at most $\lfloor n/2 \rfloor$ nodes.

Input

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output

Print one integer: a centroid node. If there are several possibilities, you can choose any of them.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5
1 2
2 3
3 4
3 5
```

Output:

```
3
```